# Stablizatoare de tensiune

# Intro

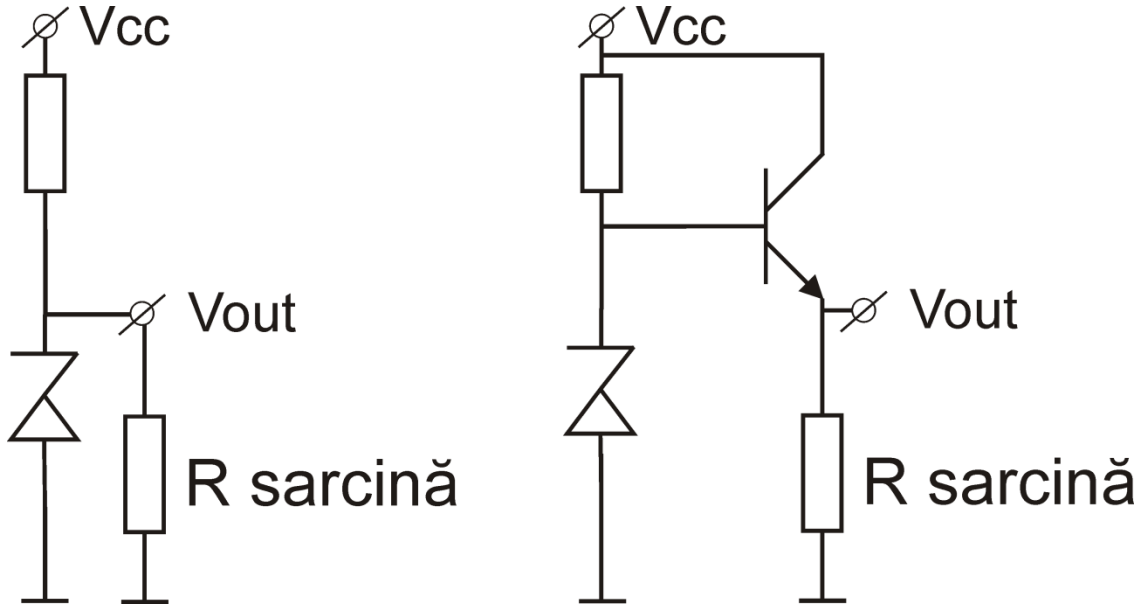
# Zenner + R / Zenner + R + TBIP





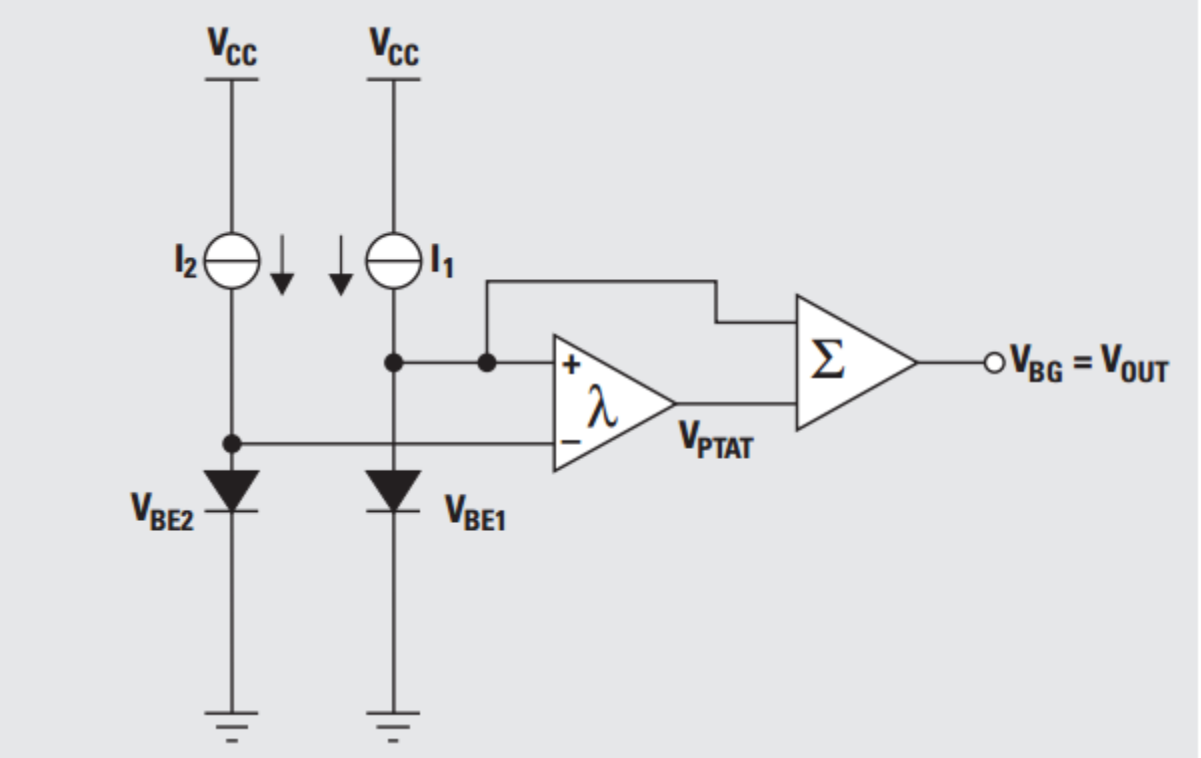

//discutie la curs
//vin vs vout
//ripple
//diff

# Zenner + R / Zenner + R + TBIP

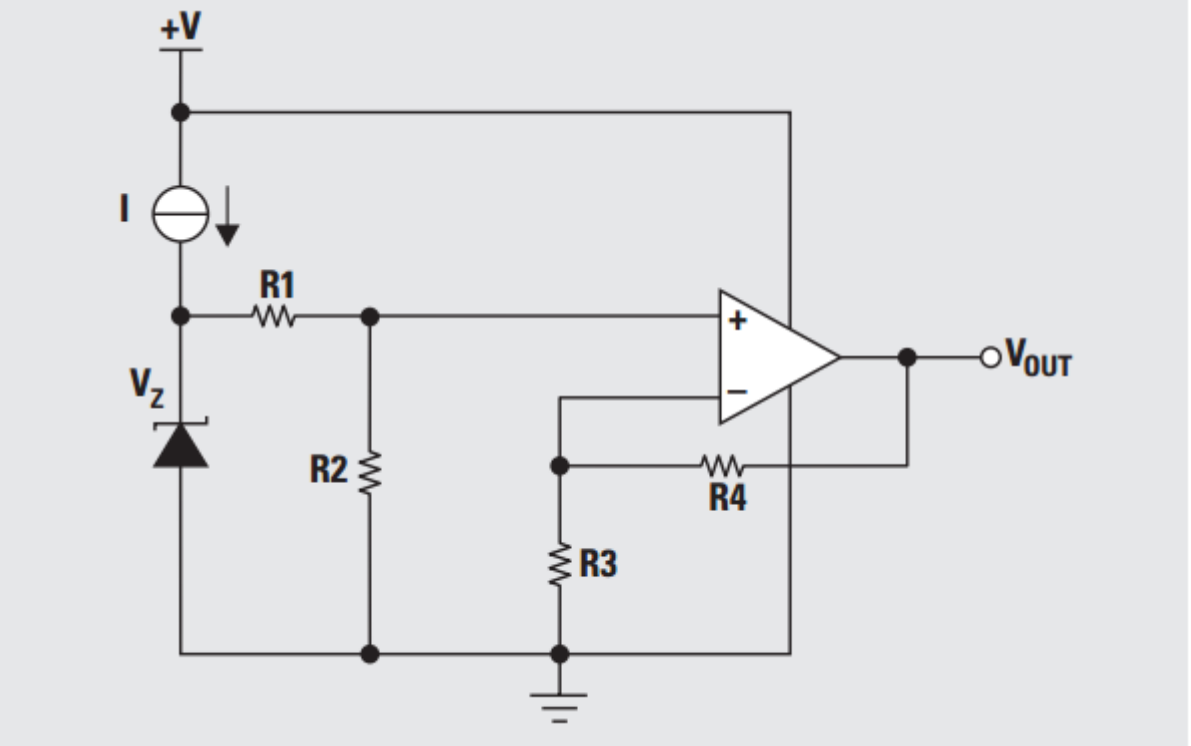


# Referințe

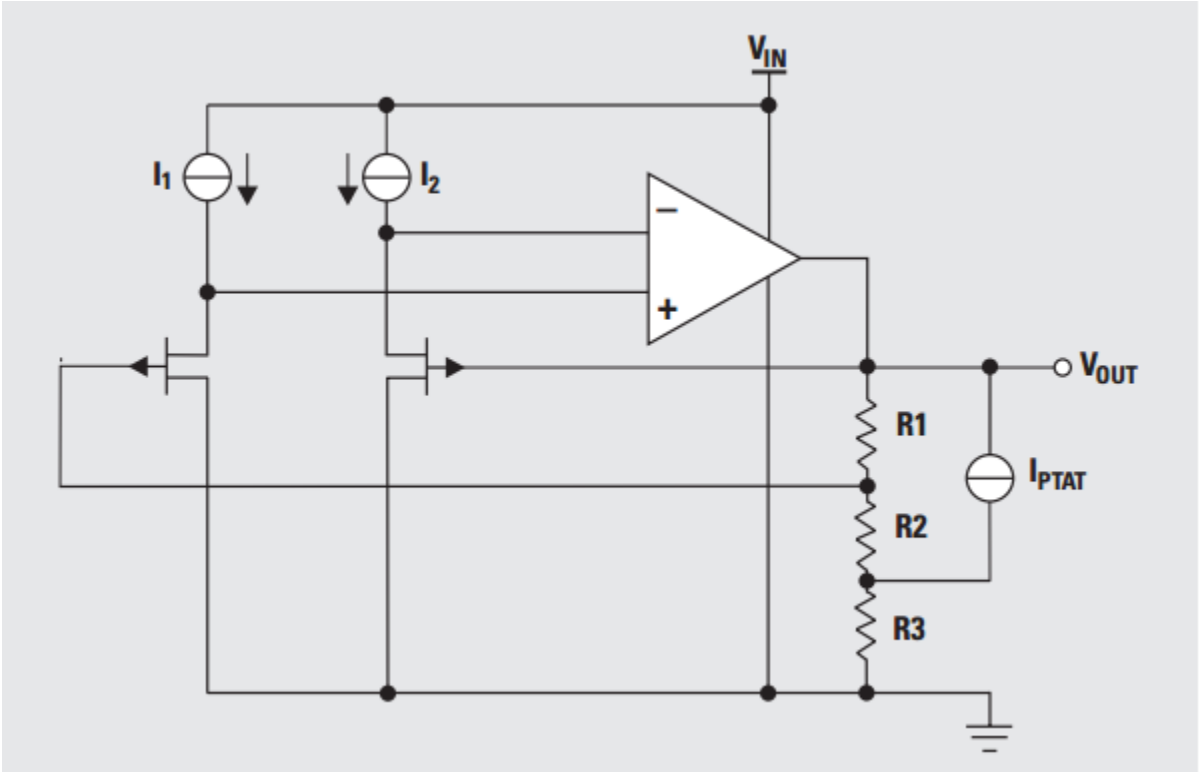
# Band Gap



# Buried zener



# XFET



# Aplicații Diverse

- Precision Reference – LM329 - http://www.ti.com/lit/ds/symlink/lm329.pdf

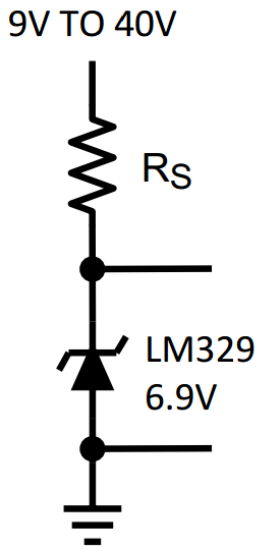
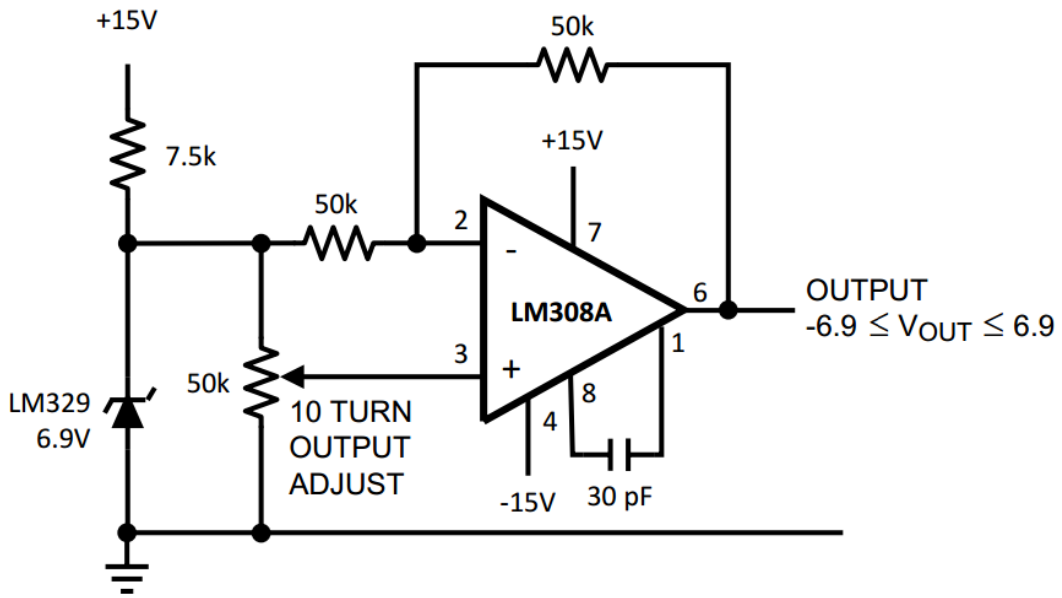



**Figure 2. Simple Reference**

# Aplicații Diverse

- Precision Reference – LM329 - http://www.ti.com/lit/ds/symlink/lm329.pdf

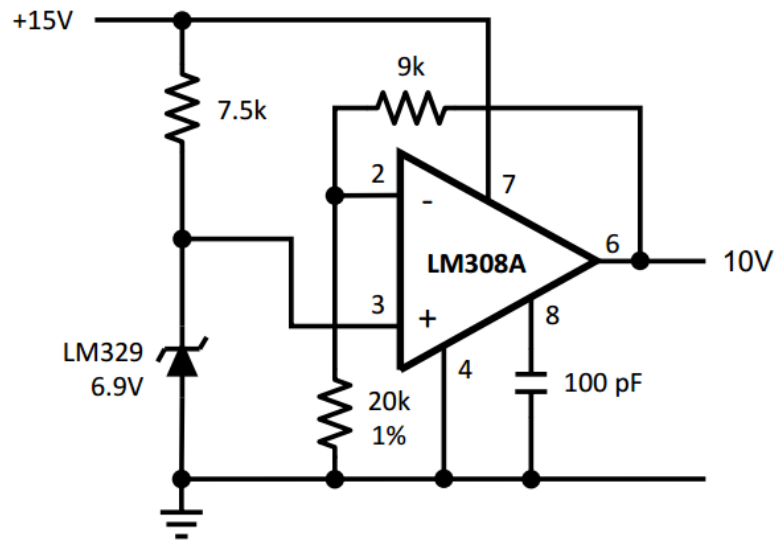
**Figure 4. Adjustable Bipolar Output Reference**



# Aplicații Diverse

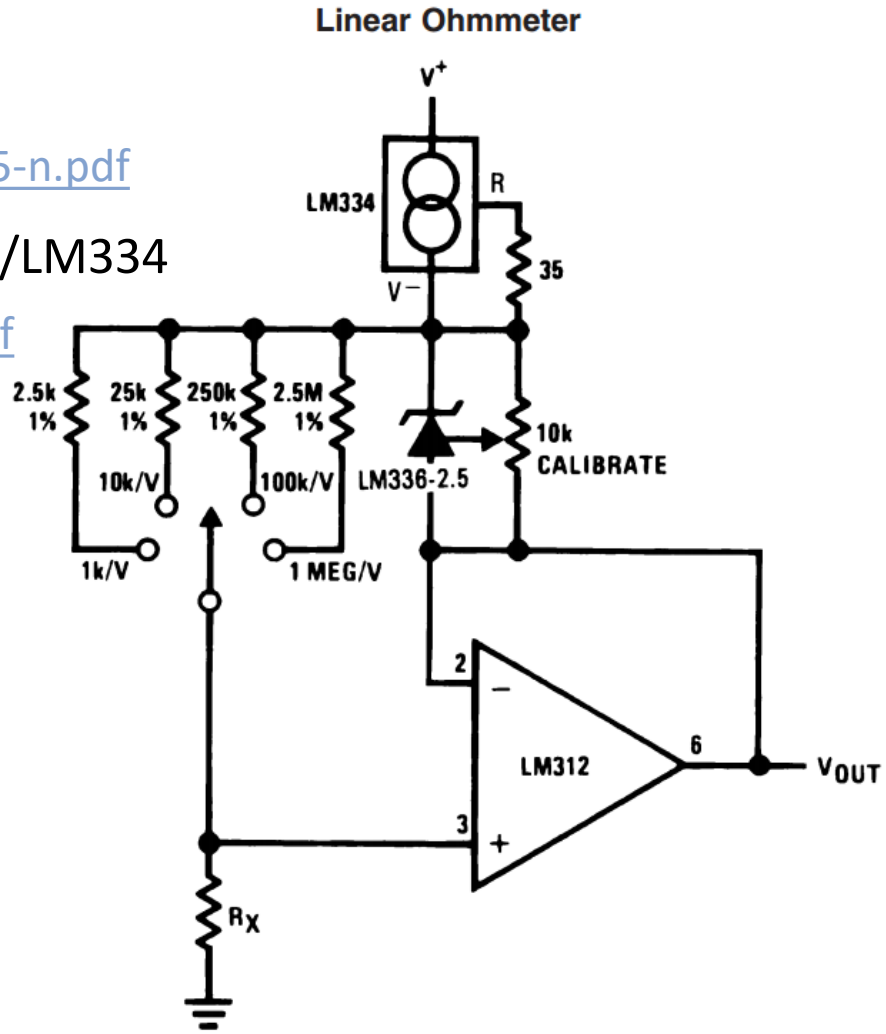
- Precision Reference – LM329 - http://www.ti.com/lit/ds/symlink/lm329.pdf

**Figure 8. Buffered Reference with Single Supply**

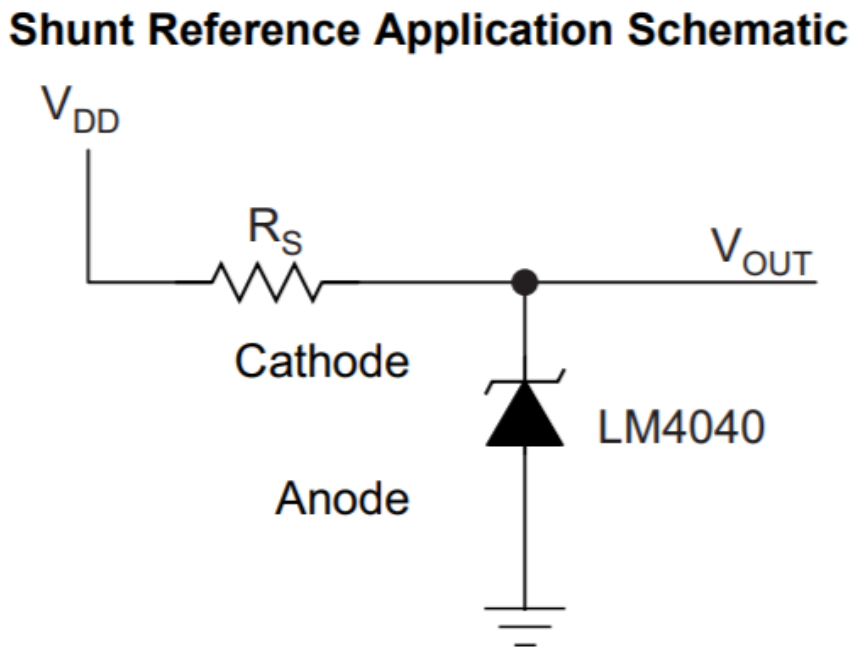


# Aplicații Diverse

- Precision Reference – LM336
  - http://www.ti.com/lit/ds/symlink/lm336-2.5-n.pdf
- Adjustabel Current Source – LM134/LM234/LM334
  - http://www.ti.com/lit/ds/symlink/lm134.pdf

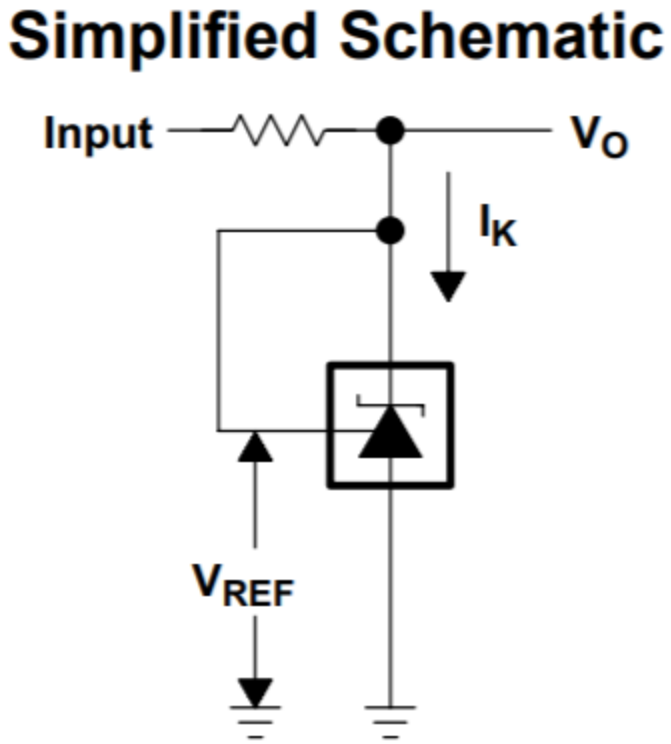


# Aplicații Diverse



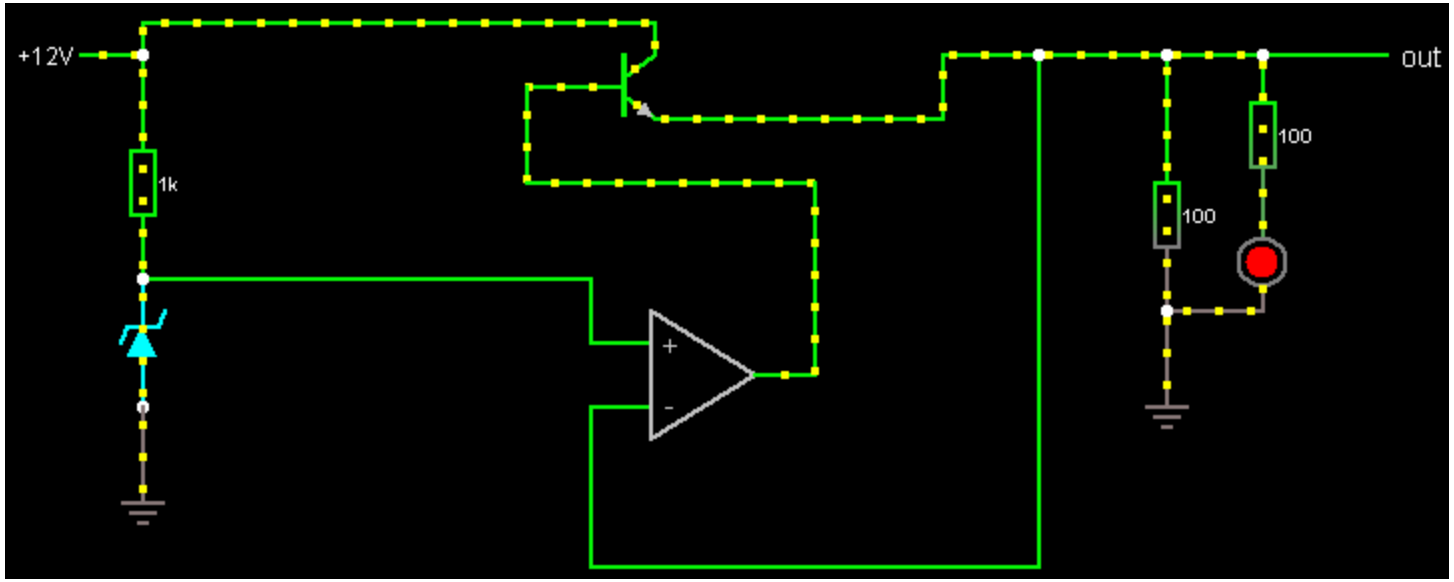

https://www.ti.com/lit/ds/symlink/lm4040-n-q1.pdf

# Aplicații Diverse

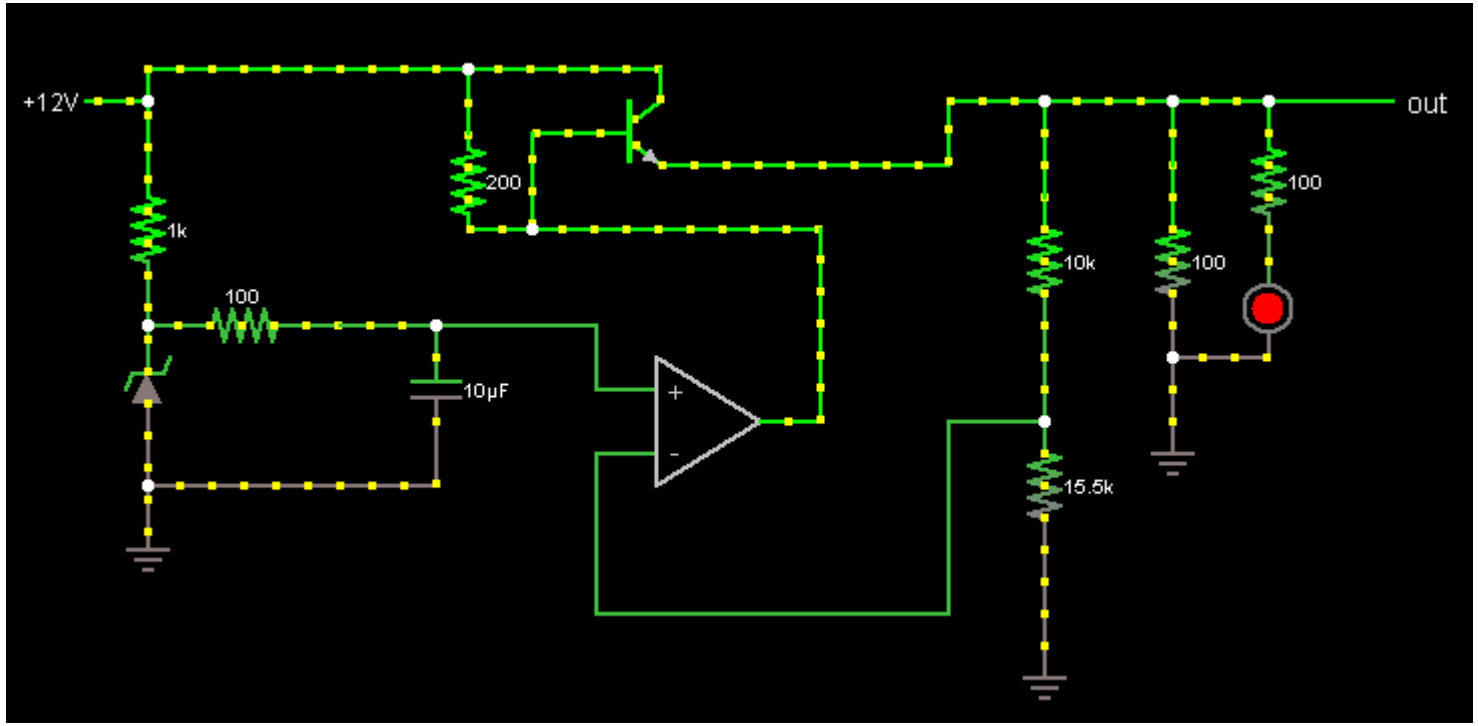



https://www.ti.com/lit/ds/symlink/tlv431.pdf

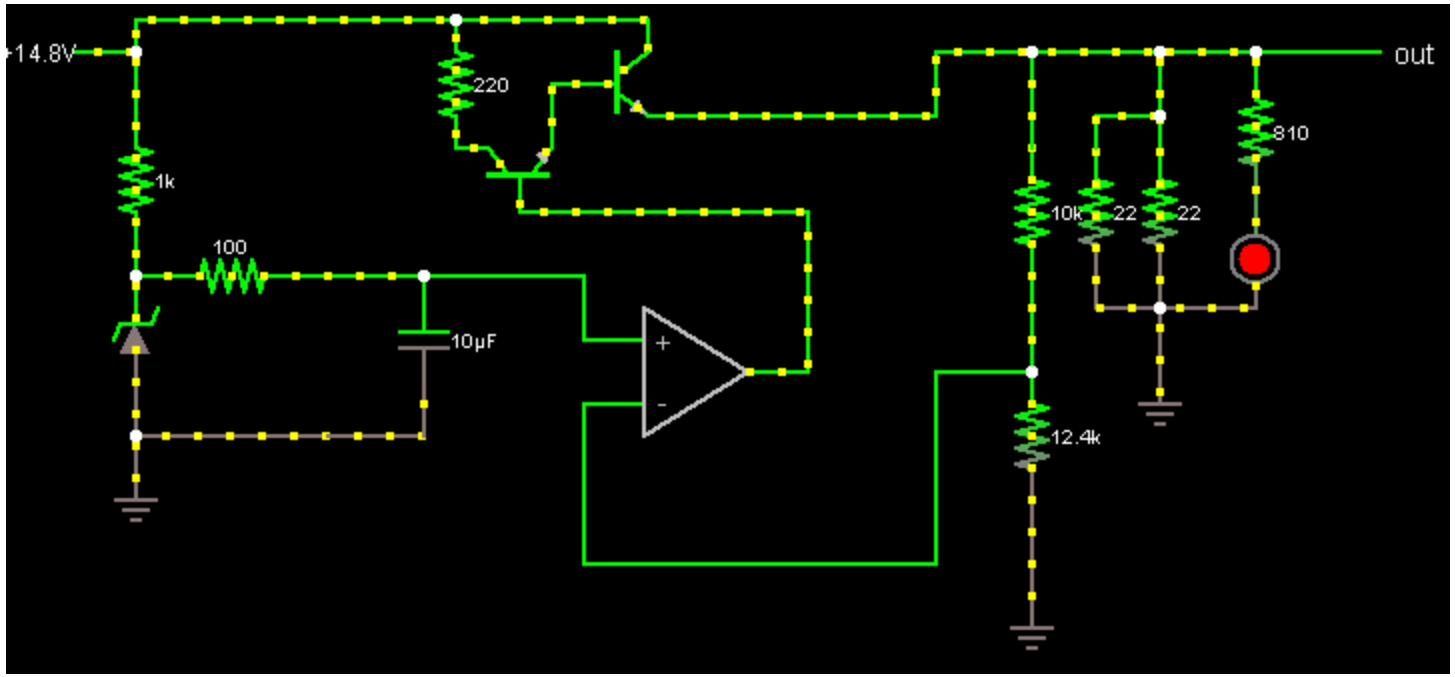
# Stabilizator de tensiune liniar (schema simplificata) – var 1



# Stabilizator de tensiune liniar (schema simplificata) – var 1

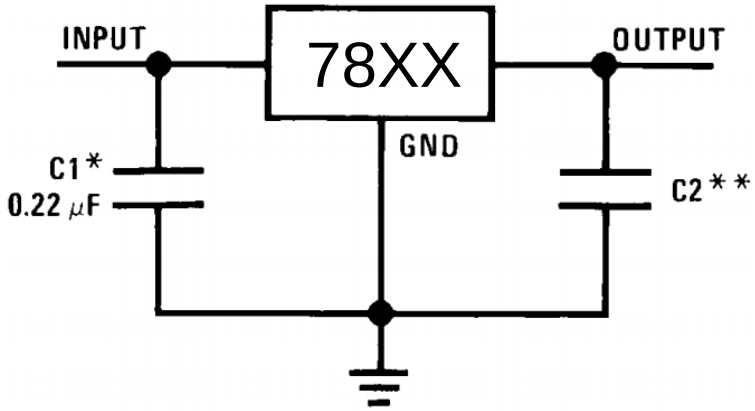


# Stabilizator de tensiune liniar (schema simplificata) – var 3



# Stablizatoare de tensiune – liniare integrate

# Liniar Voltage regulators – familia 78




//discutie la curs
//output voltage
//c-in
//c-out

//discutie la curs:
- Dropout voltage
- Quiescent current
- Line regulation
- Load regulation

© https://www.ti.com/lit/ds/symlink/lm7800.pdf?ts=1610533958575

# Liniar Voltage regulators – familia 78

Tensiuni pozitive:

- LM78xx - http://www.fairchildsemi.com/ds/LM/LM7805.pdf
- uA78xx – http://www.ti.com/lit/ds/symlink/ua7805.pdf
- uA78lxx - http://www.datasheetcatalog.org/datasheets/208/402911_DS.pdf

Tensiuni negative:

- LM79xx - http://www.fairchildsemi.com/ds/LM/LM7905.pdf

# Param de discutat

(la curs)

- Voltage dropout
- Quiescent current
- Current limit
- PSSR (Power-supply rejection rat $PSRR(dB) = 20 \log \frac{V_{ripple(in)}}{V_{ripple(out)}}$
- Output noise

# Alte exemple

# 78xx - aplicatii

- https://www.fairchildsemi.com/datasheets/LM/LM7805.pdf

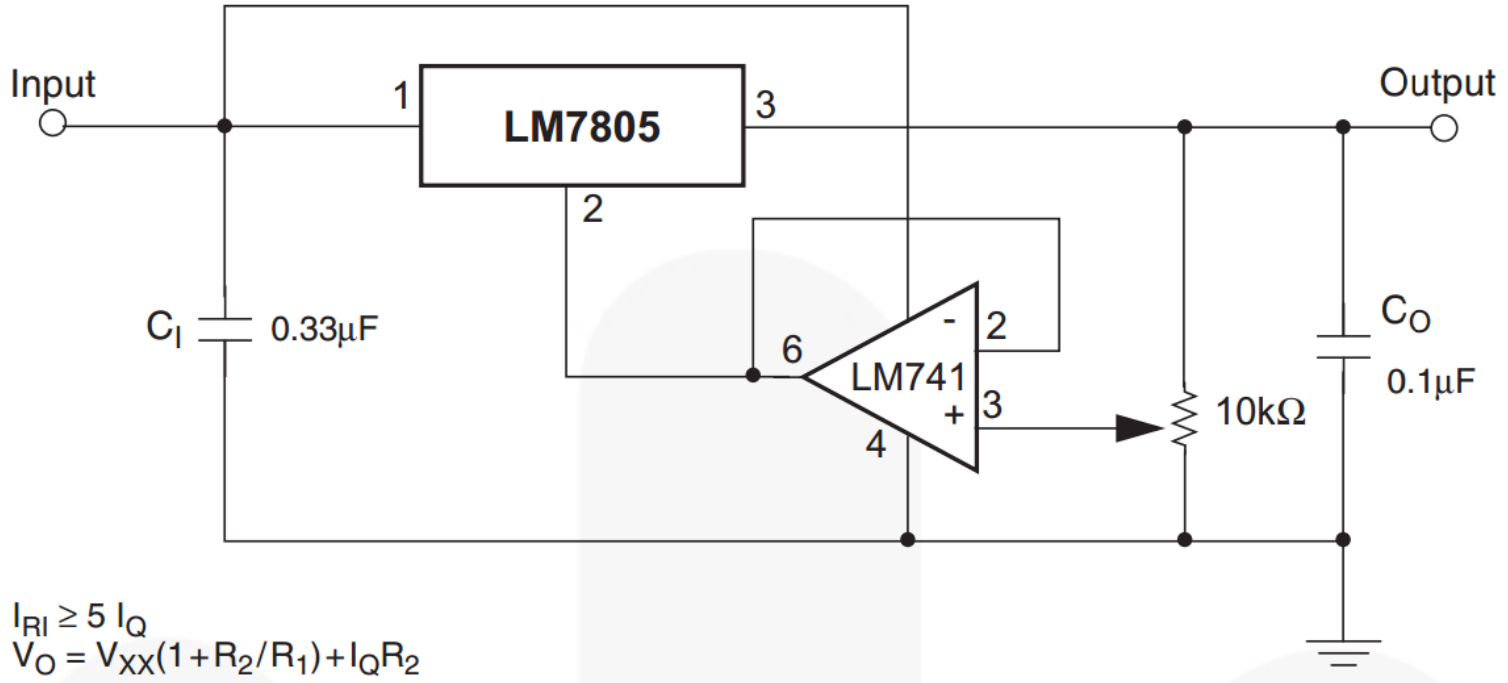



**Figure 12. Adjustable Output Regulator (7 V to 30 V)**

# 78xx - aplicatii

- https://www.fairchildsemi.com/datasheets/LM/LM7805.pdf

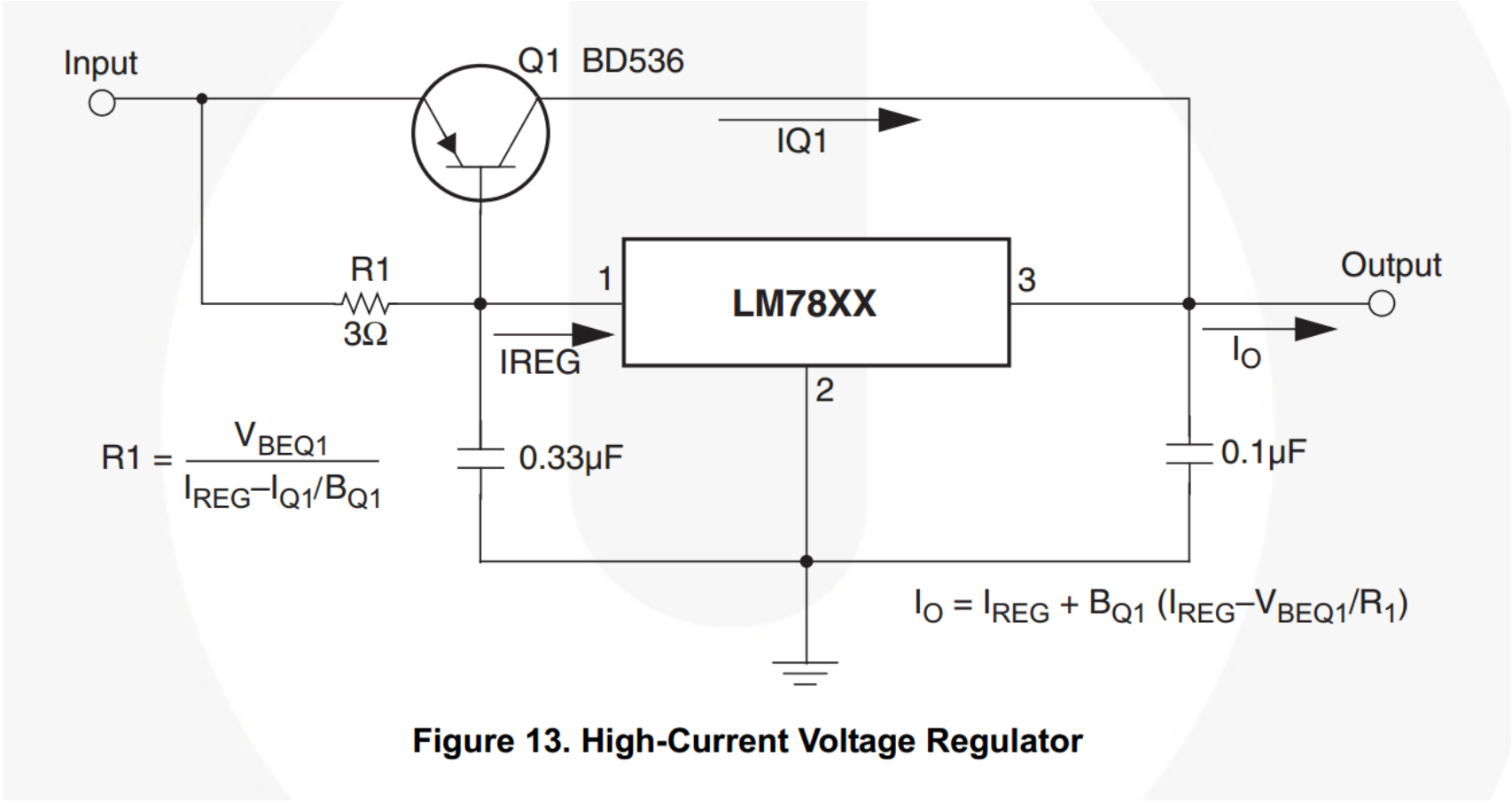



**Figure 13. High-Current Voltage Regulator**

# 78xx - aplicatii

- https://www.fairchildsemi.com/datasheets/LM/LM7805.pdf

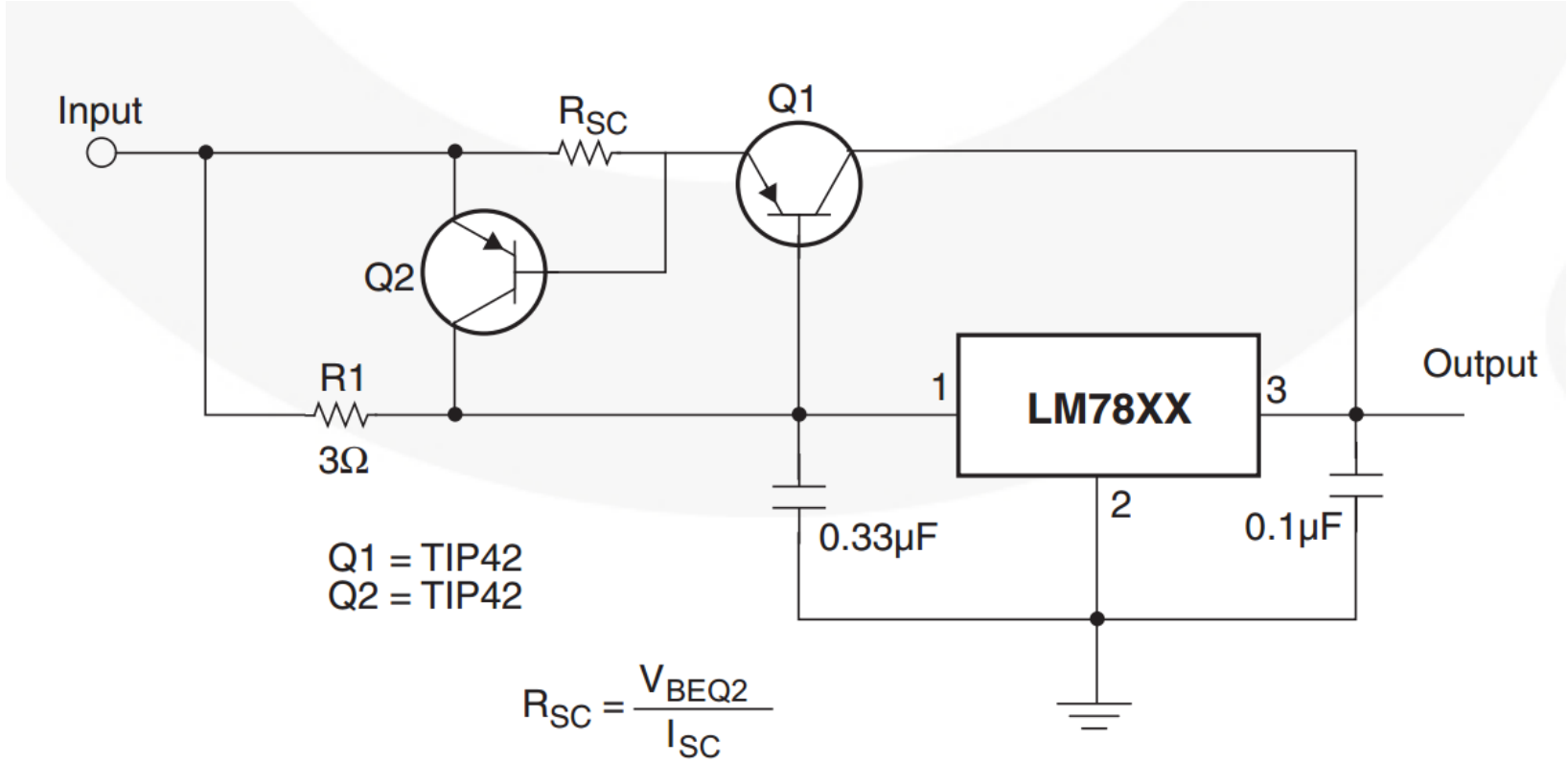



**Figure 14. High Output Current with Short-Circuit Protection**

# 78xx - aplicatii

- https://www.fairchildsemi.com/datasheets/LM/LM7805.pdf

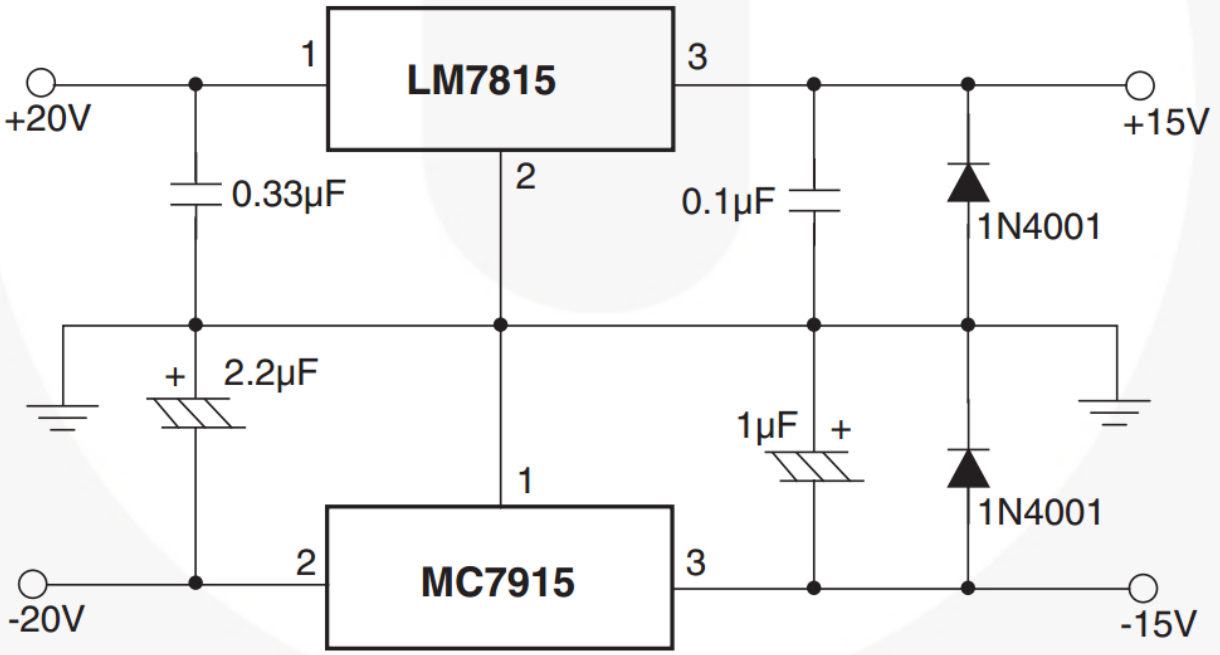
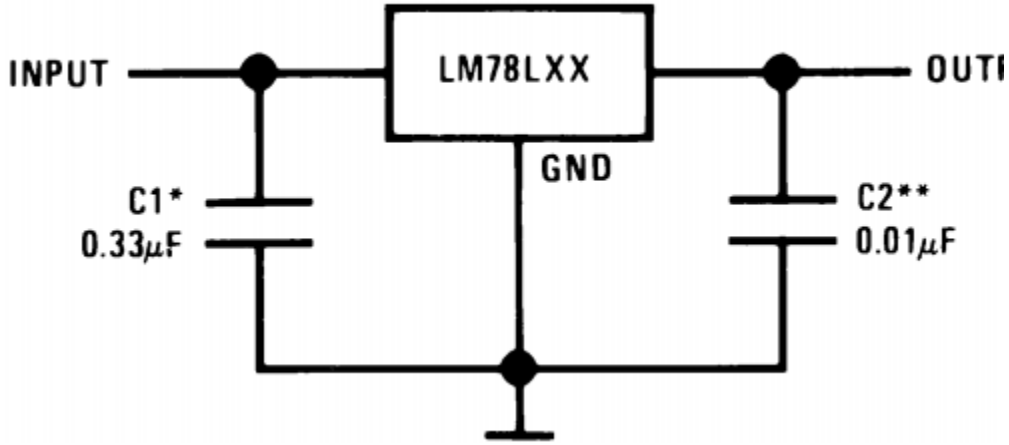



**Figure 16. Split Power Supply (±15 V - 1 A)**

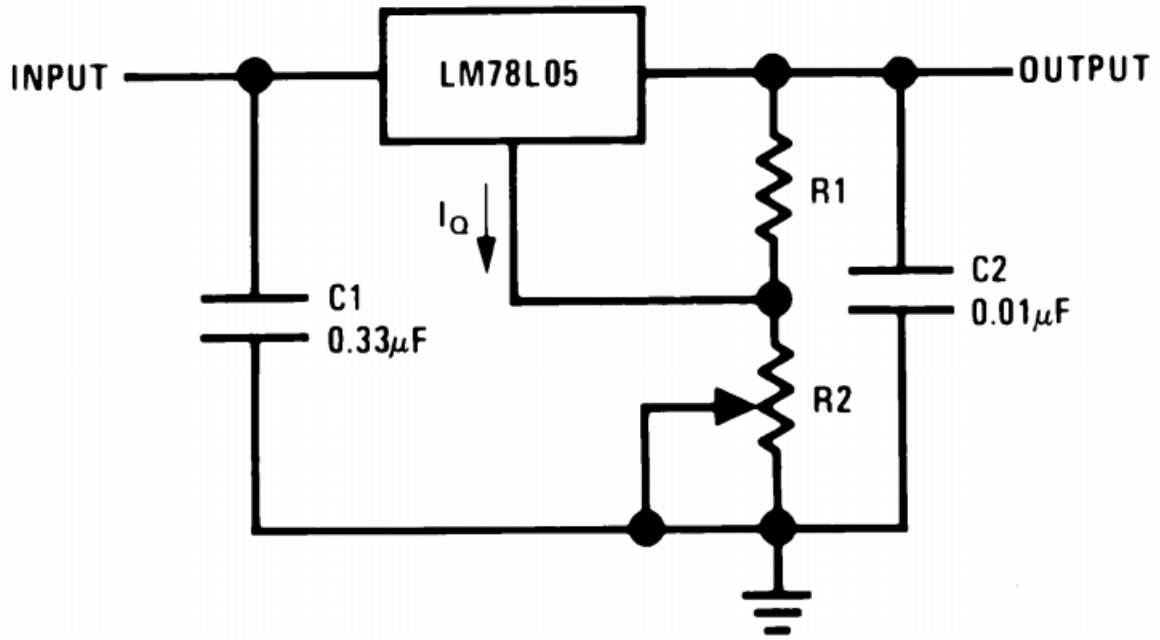
# Liniar Voltage regulators – familia 78lxx

- http://www.ti.com/lit/ds/symlink/lm78l05.pdf
- 100mA



# Liniar Voltage regulators – familia 78lxx

- http://www.ti.com/lit/ds/symlink/lm78l05.pdf




$V_{OUT} = 5V + (5V/R1 + I_Q) R2$

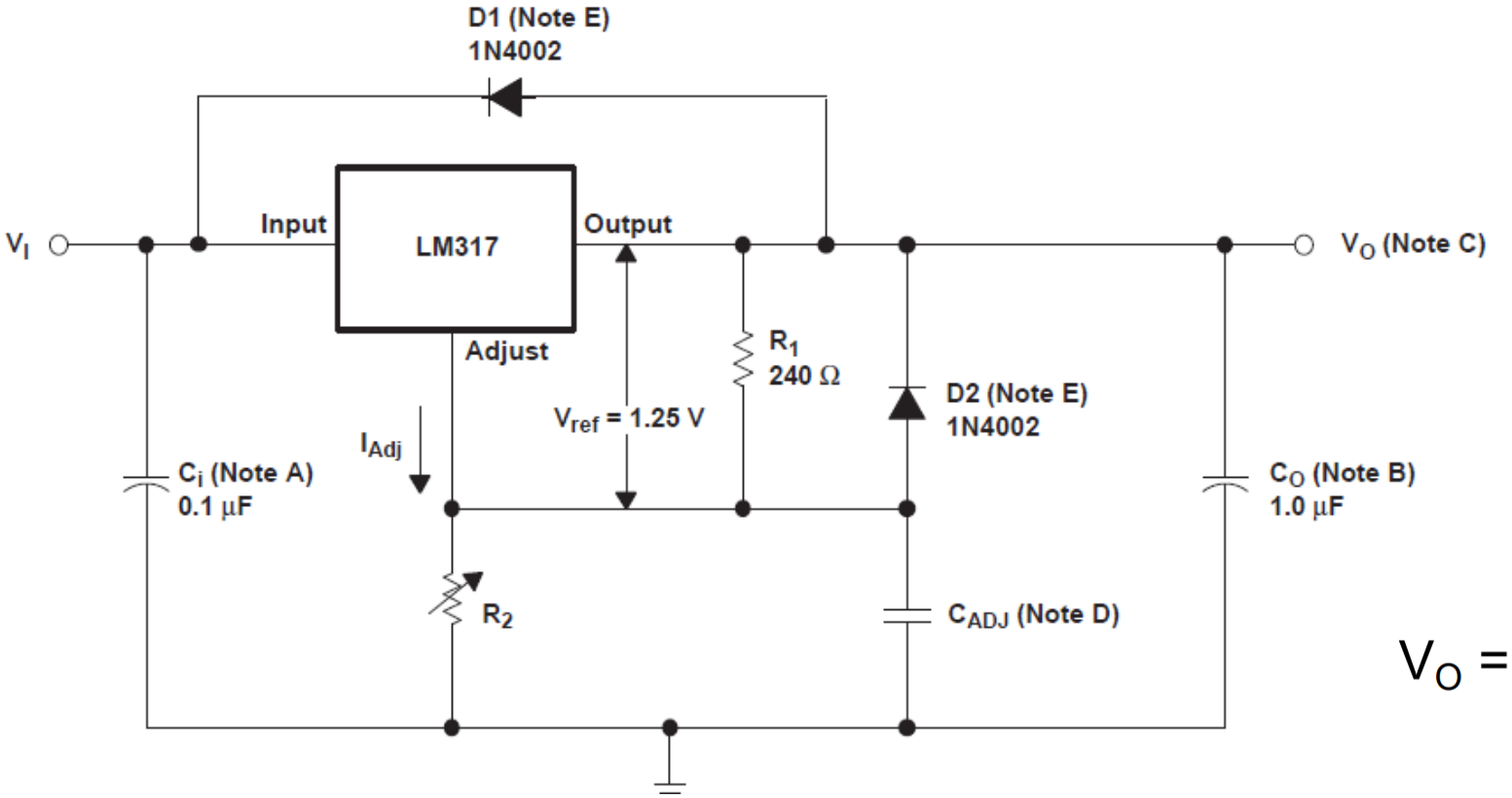
$5V/R1 > 3\ I_Q$, load regulation $(L_r) \approx [(R1 + R2)/R1]$ ($L_r$ of LM78L05)

**Figure 14. Adjustable Output Regulator**

# Liniar Voltage regulators – LM317

- http://www.ti.com/lit/ds/symlink/lm317.pdf

**(pag7)**




$$V_O = V_{REF} (1 + R2 / R1) + (I_{ADJ} \times R2)$$

# LM317 - aplicatii

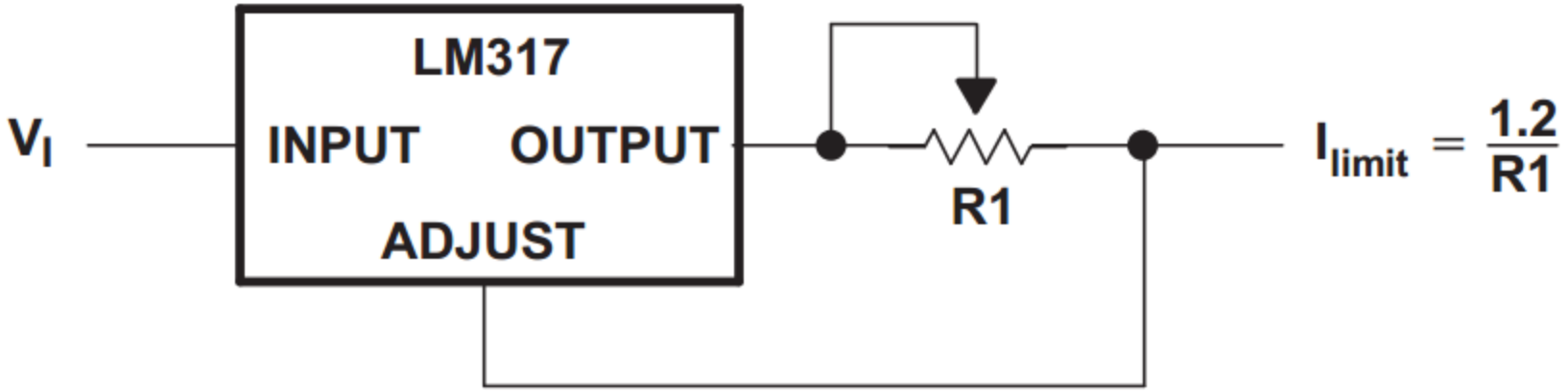



**Figure 4. Precision Current-Limiter Circuit**

# LM317 - aplicatii

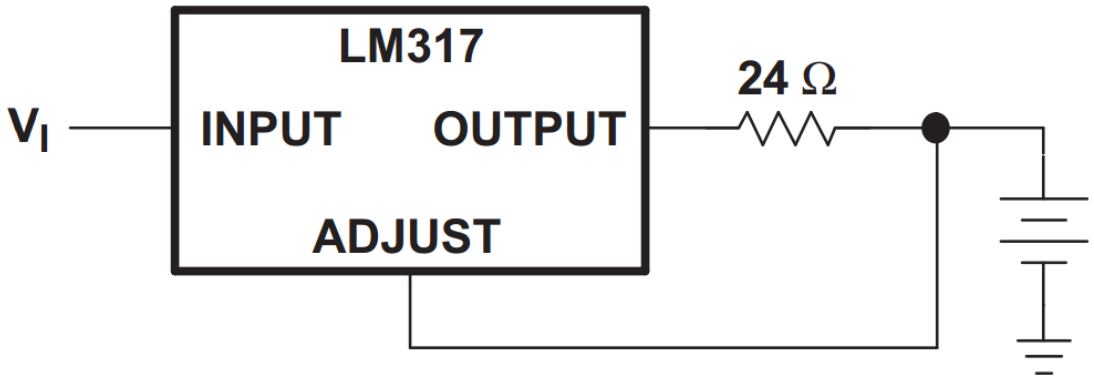


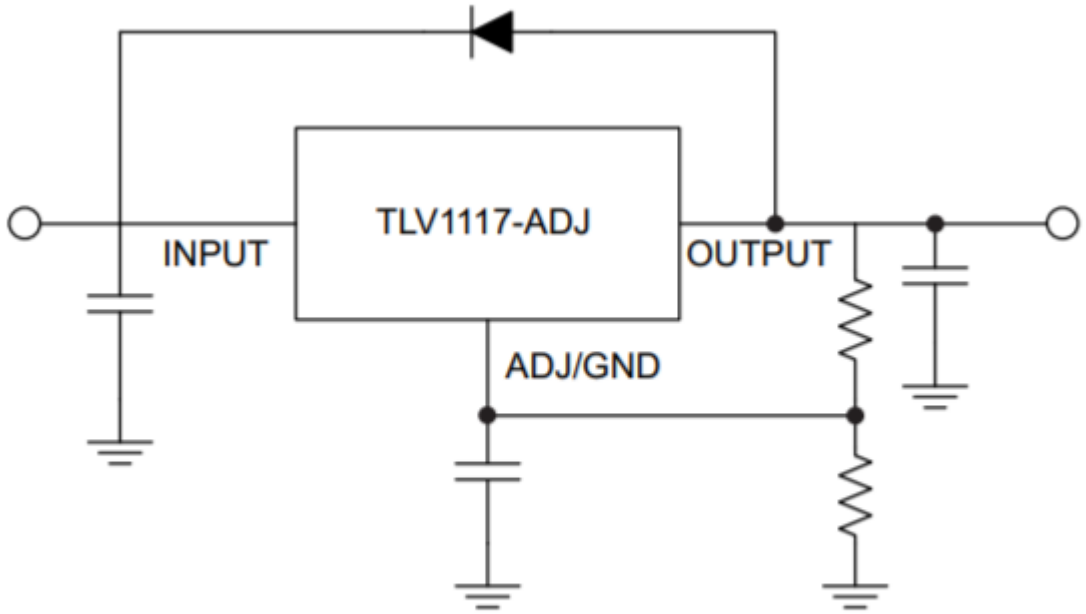

**Figure 9. 50-mA Constant-Current Battery-Charger Circuit**

# LDO - Low Dropout

# Motivatie

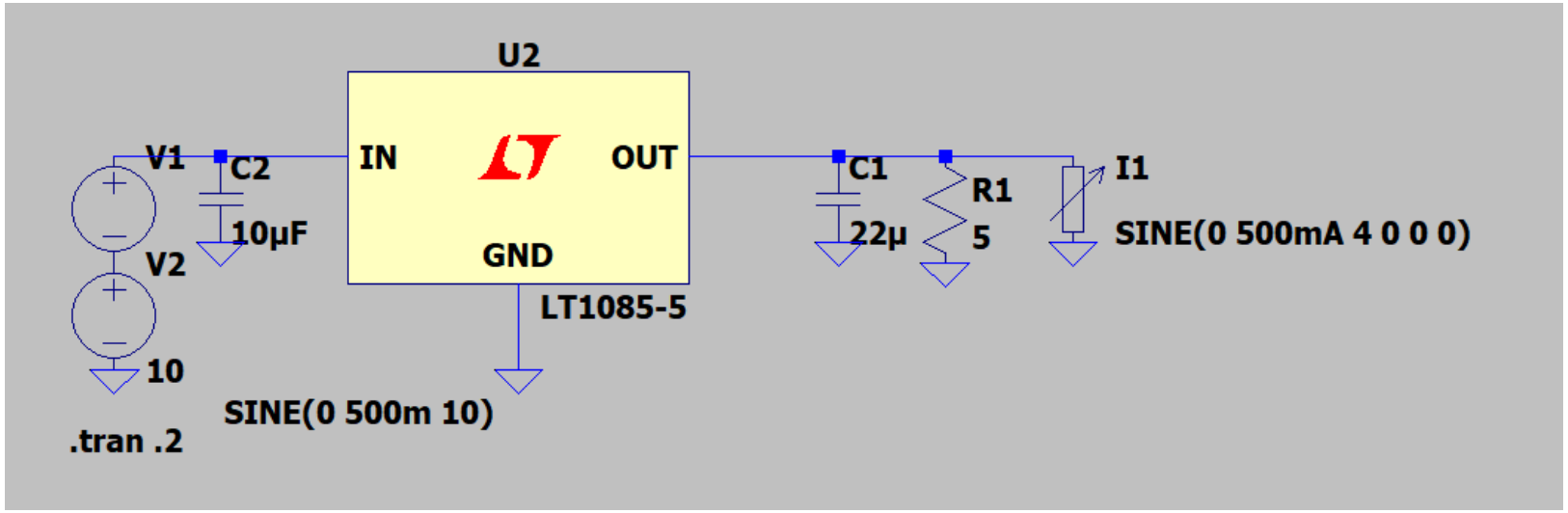
discutie Vout vs Vin

# TLV1117

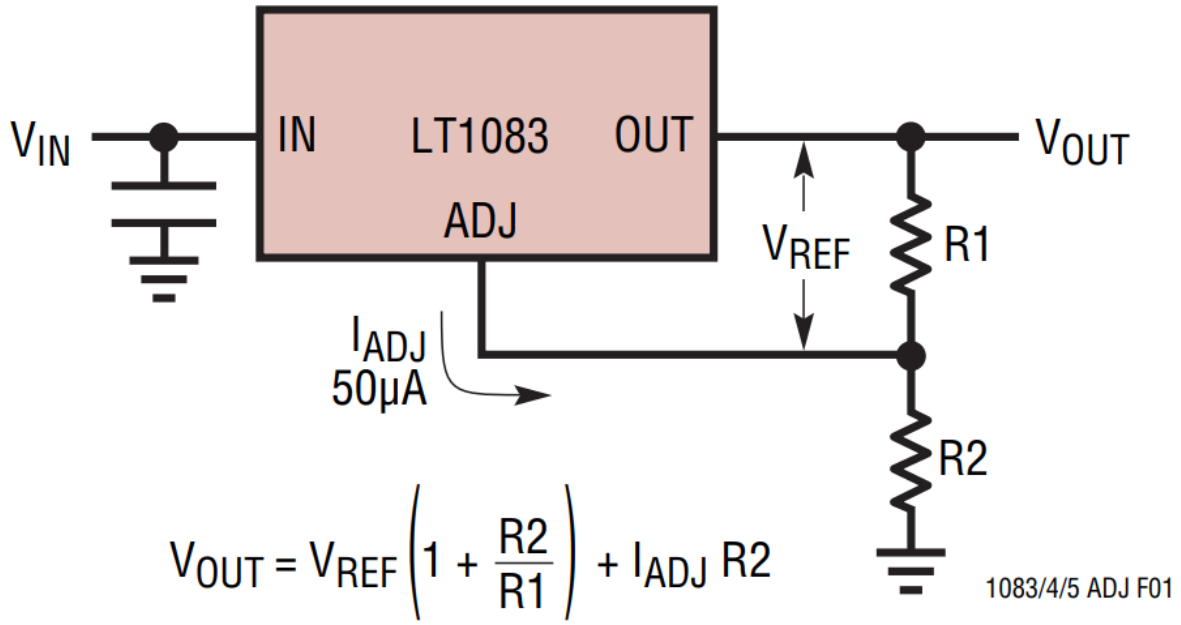



https://www.ti.com/lit/ds/symlink/tlv1117.pdf

# Simulare


U2
V1
C2
IN
OUT
C1
R1
I1
10µF
22µ
5
SINE(0 500mA 4 0 0 0)
V2
GND
LT1085-5
10
SINE(0 500m 10)
.tran .2

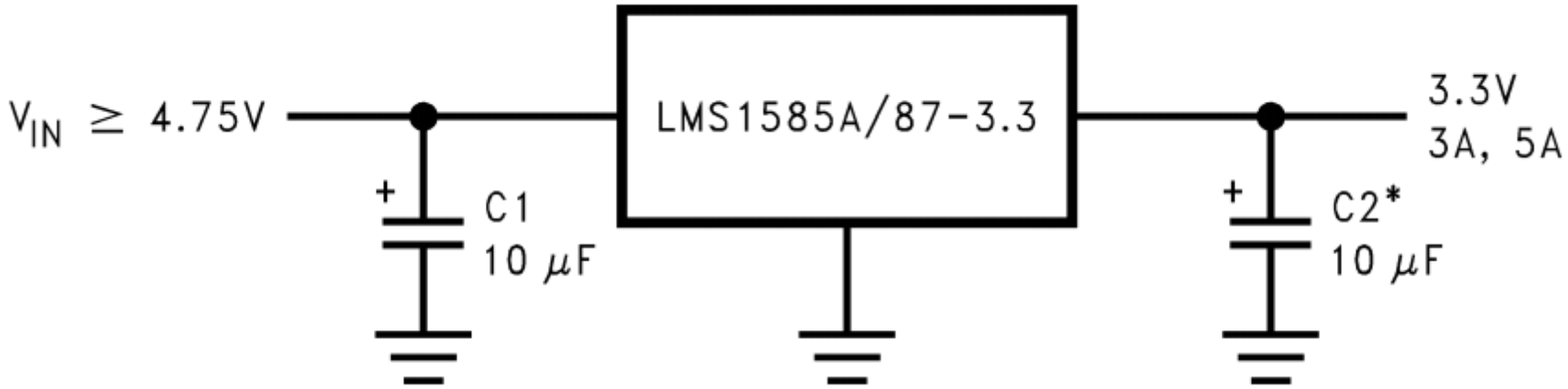
# Discutie ADJ




https://www.analog.com/media/en/technical-documentation/data-sheets/108345fh.pdf

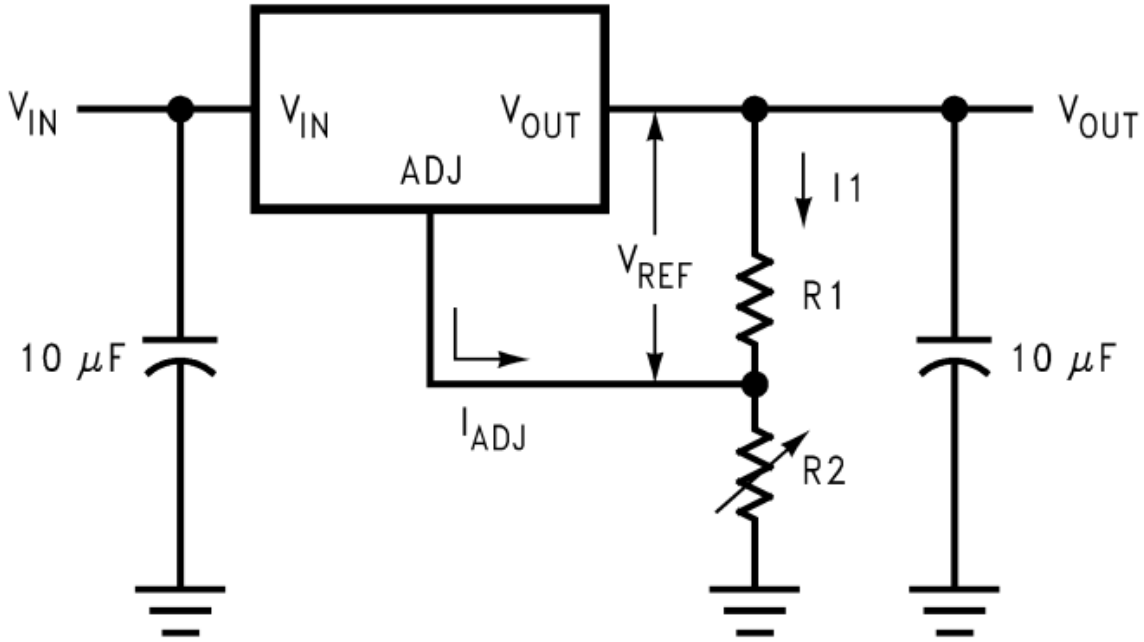
# Variante pentru curenti mari

- Tensiuni mici (<3.3V) & curenti mari (5A) http://www.farnell.com/datasheets/36807.pdf



# Variante pentru curenti mari

- Tensiuni mici (<3.3V) & curenti mari (5A) http://www.farnell.com/datasheets/36807.pdf




$$V_{OUT} = V_{REF}\left(1 + \frac{R2}{R1}\right) + I_{ADJ}\,R2$$

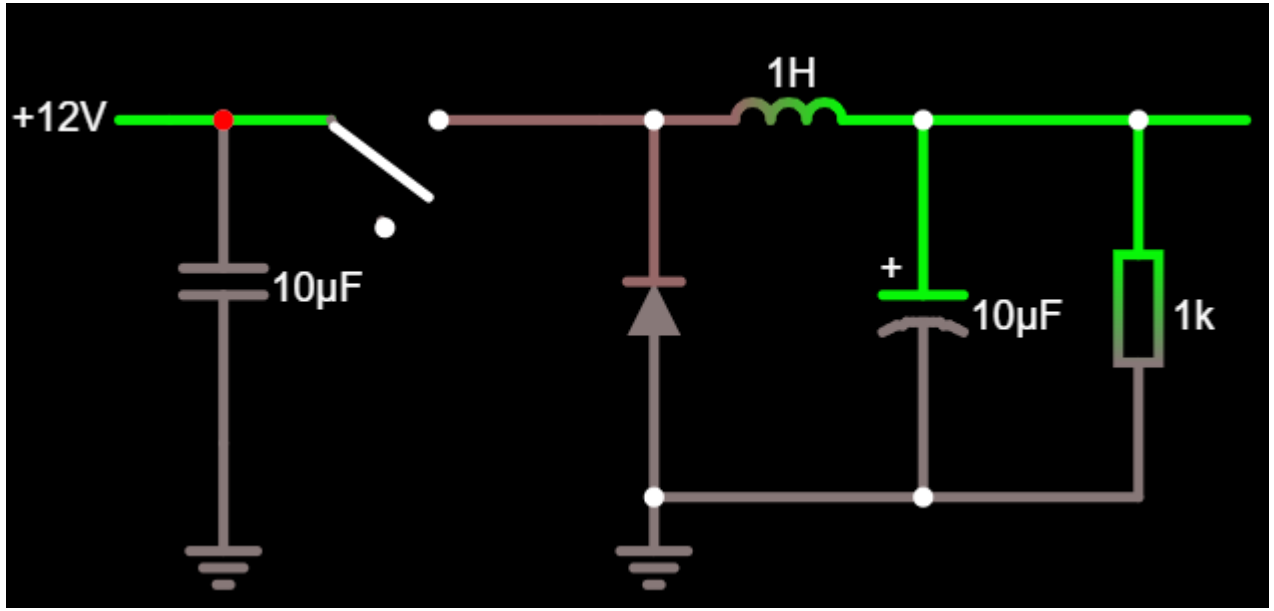
# Variante pentru curenti mici

- http://www.ti.com/lit/ds/symlink/lp3964.pdf
  - **dropout – 340mV la 800mA**

# Surse în comutație

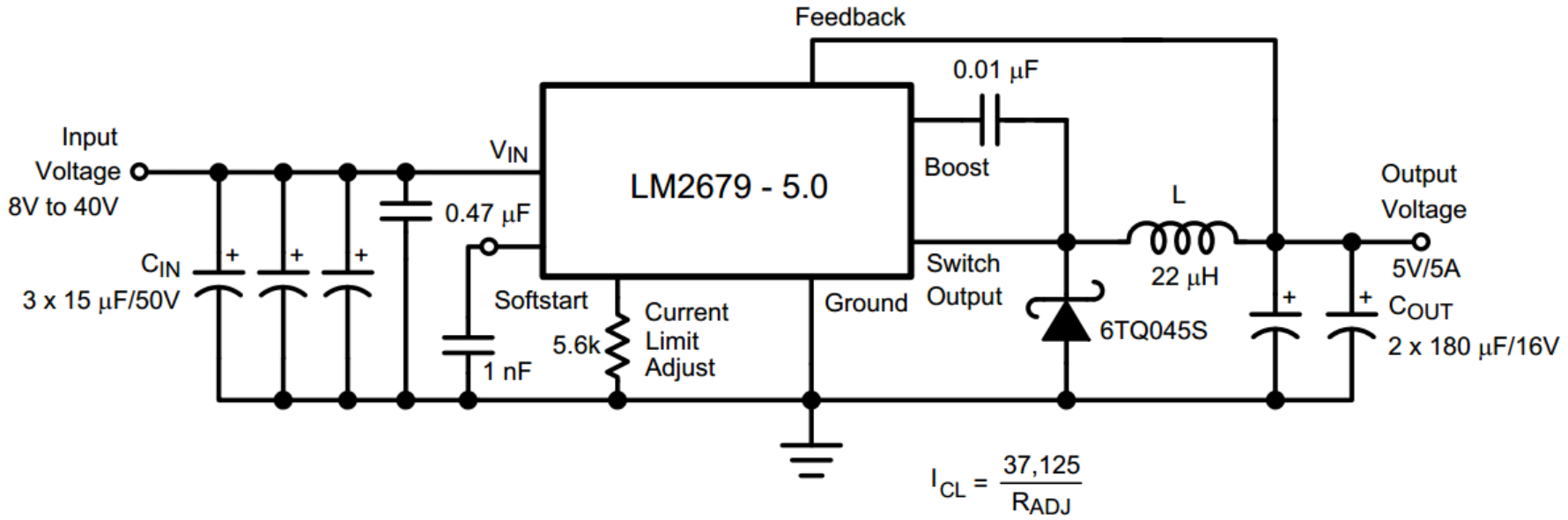
# Discutie - eficienta

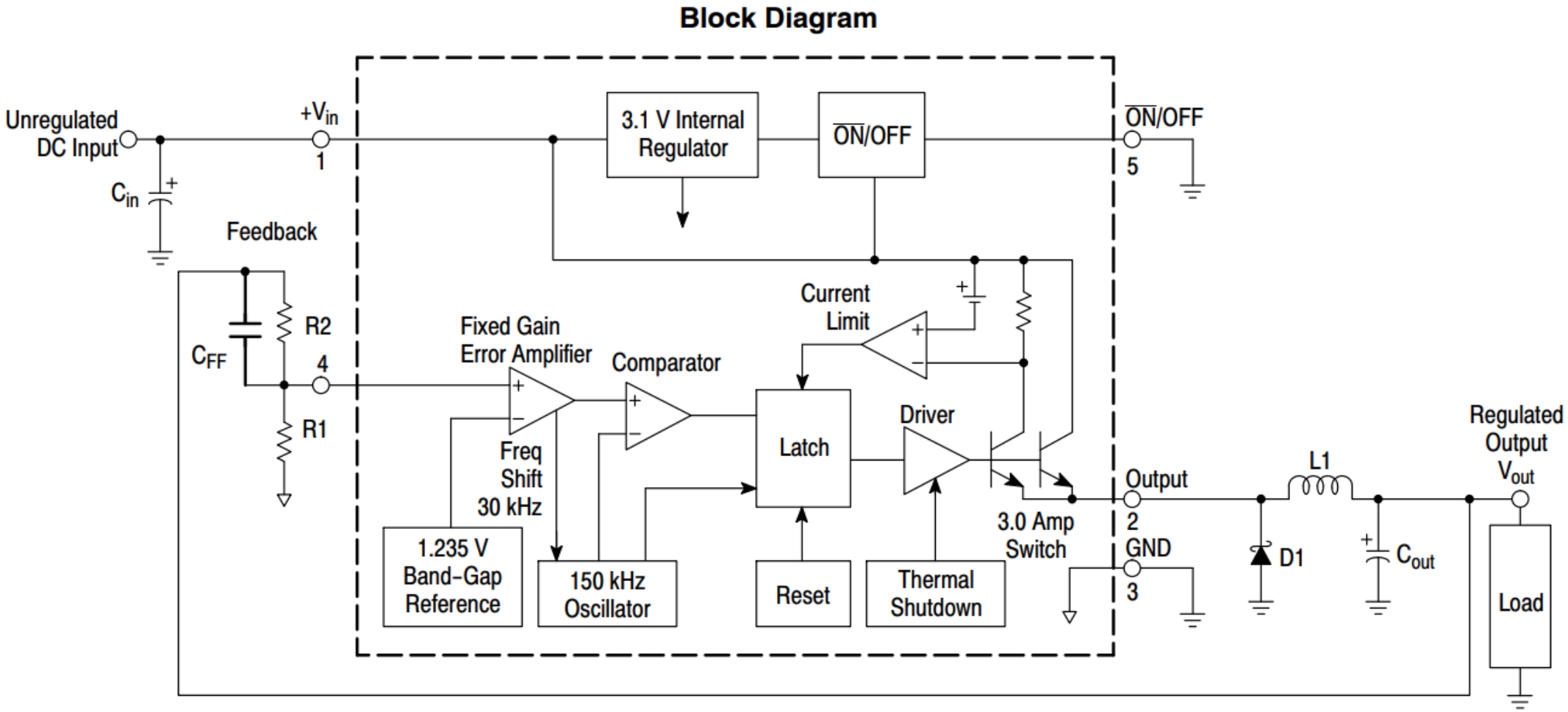
//calcul P
//discutie eficienta
//schema & explicatie  buck convertor



# LM2679

- *Discutie la curs*
- http://www.ti.com/product/lm2679






© https://www.onsemi.com/pub/Collateral/LM2596-D.PDF

# LM2679

- http://www.ti.com/product/lm2679

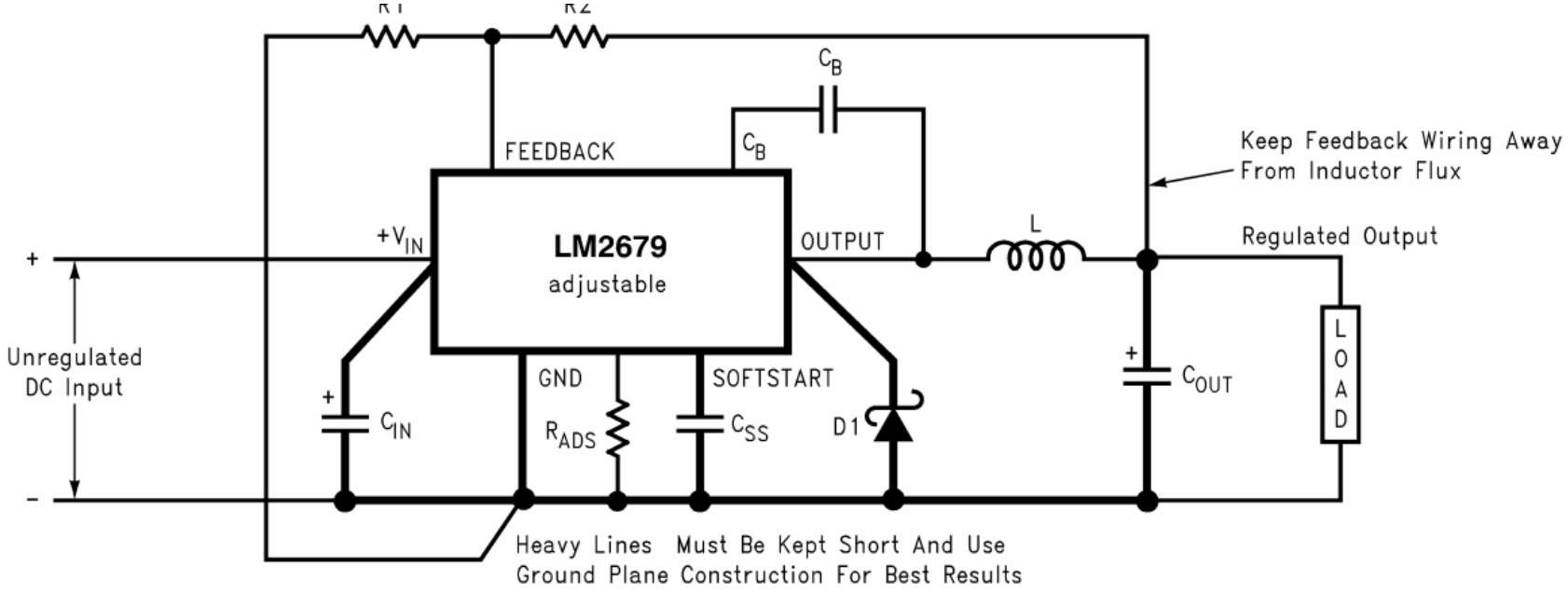
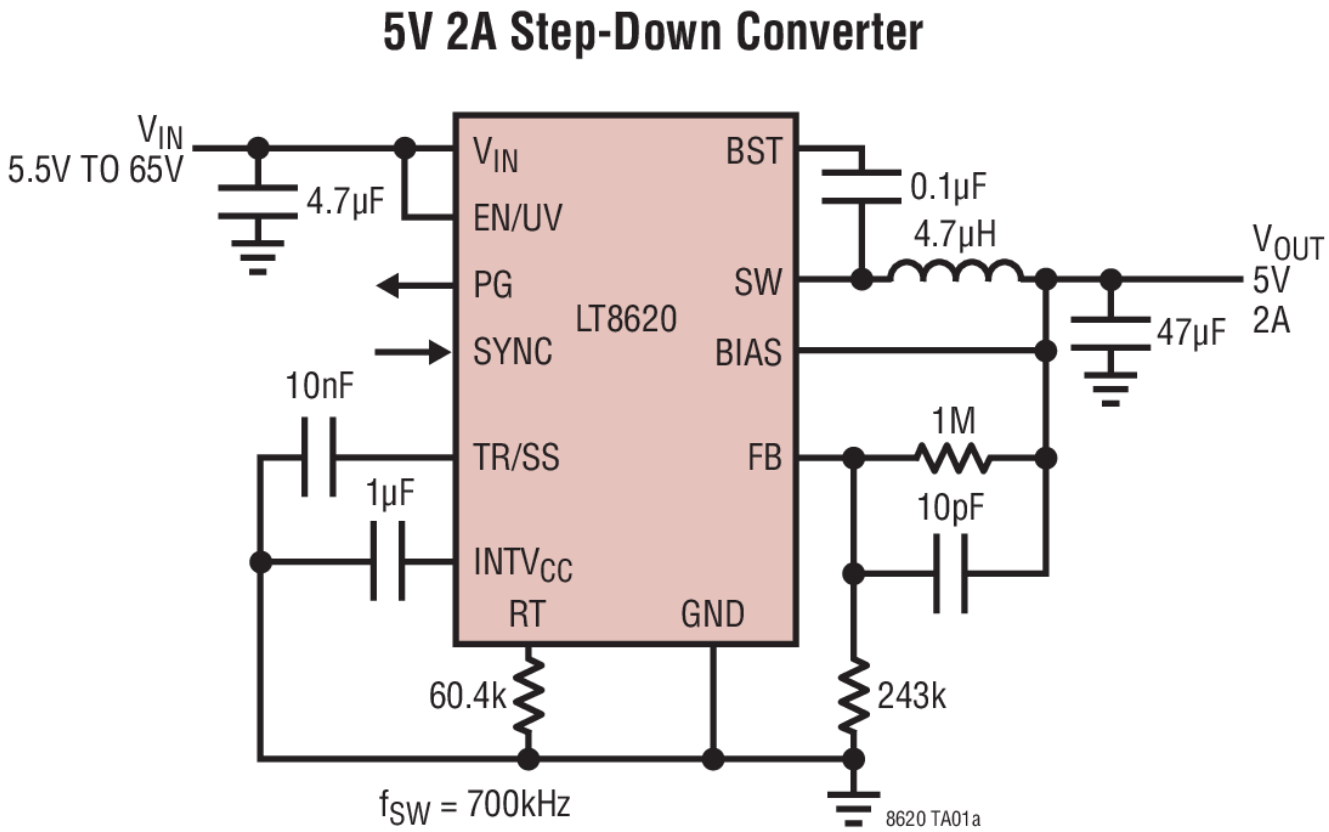



**Figure 17. Basic Circuit for Adjustable Output Voltage Applications**

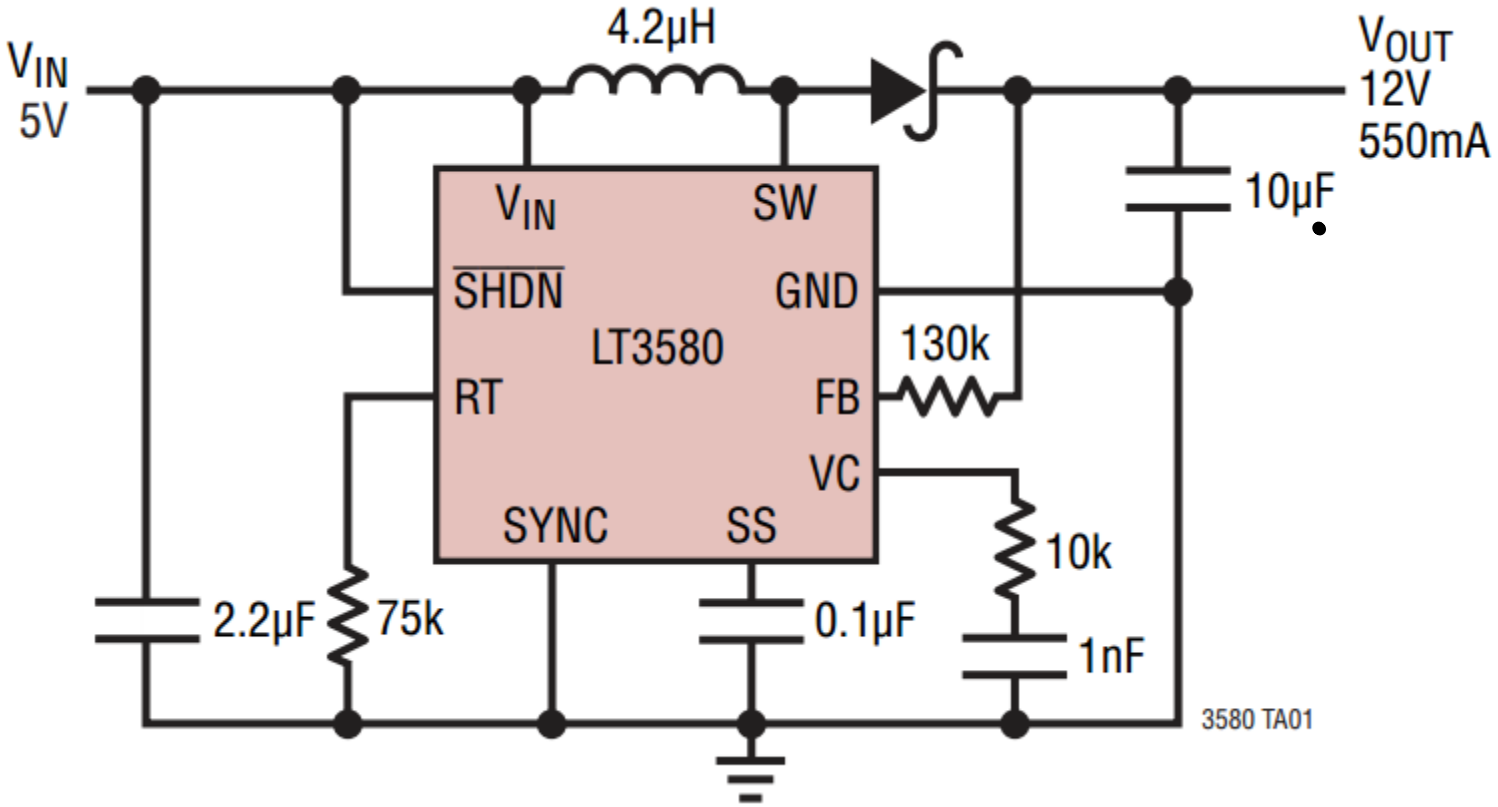
# LT8620

- http://www.linear.com/product/LT8620



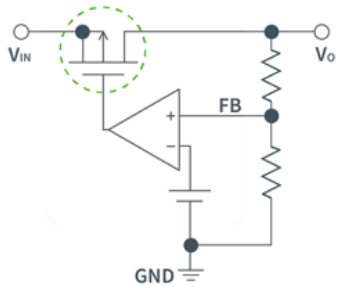
# Boost – Ex: LT3580

- *Discutie la curs*
- http://cds.linear.com/docs/en/datasheet/3580fg.pdf



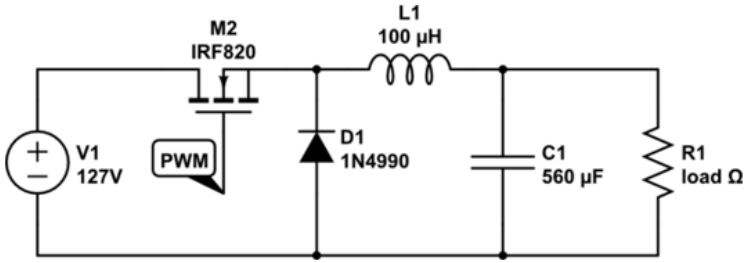
# Comparatie

## Linear Regulators




- Low Cost (low power DC applications)
- Simple to design
- High PSRR (low noise)
- Low efficiency if $V_O \ll V_I$
- Bigger heat losses than SMPS

## Switch Mode Power Supplies

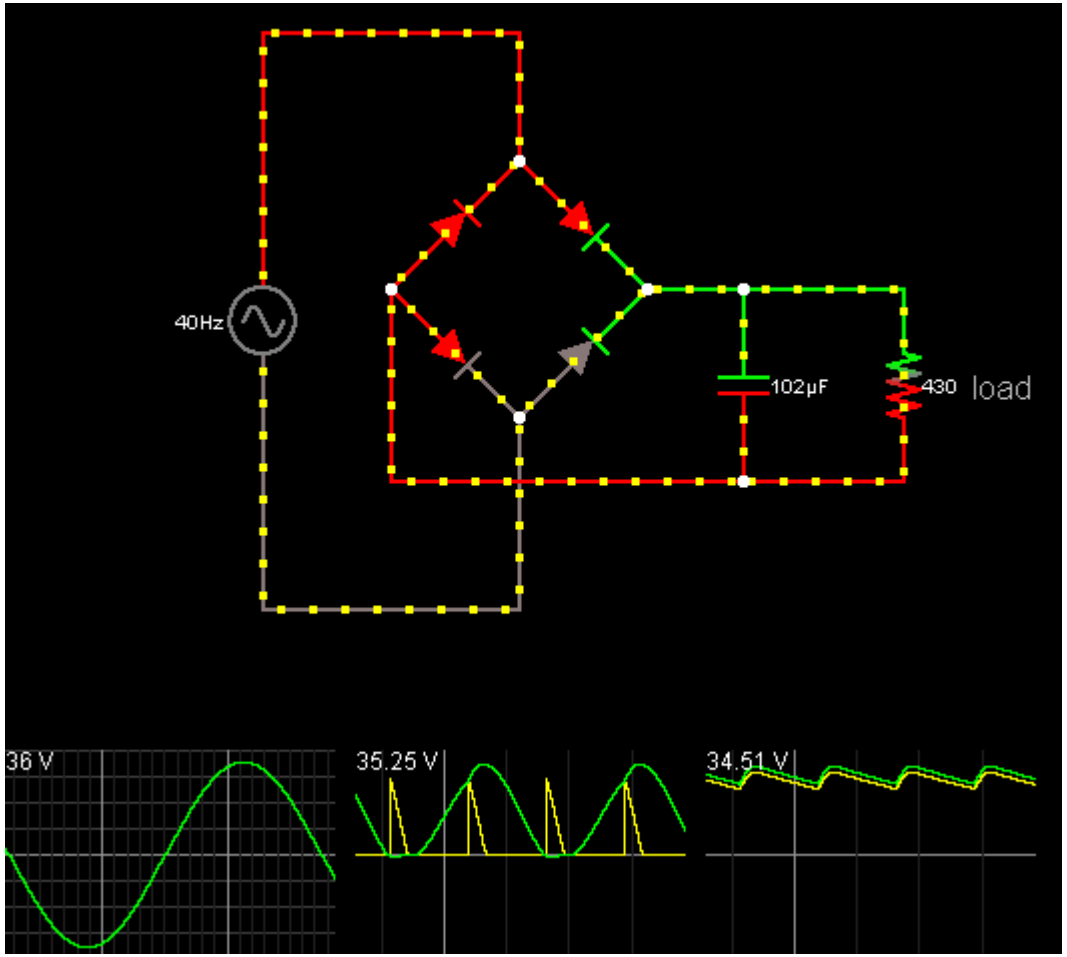



- Higher efficiency
- More W/cm$^3$
- Lower cost than linear supplies when used in AC-DC conversion (especially for high power)
- Source of EMI and can cause EMC problems
- Complex design (requires skilled engineer)
- Slower transient response

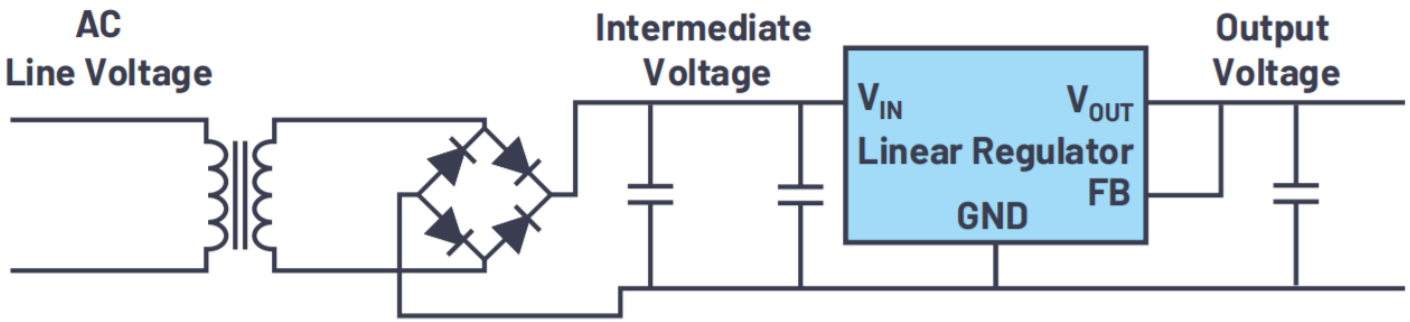
*© prezentare Switch Mode Power Supplies @ Alexandru Aungurencei*

# AC main > low DC

# Puntea redresoare



# Surse AC – DC liniare

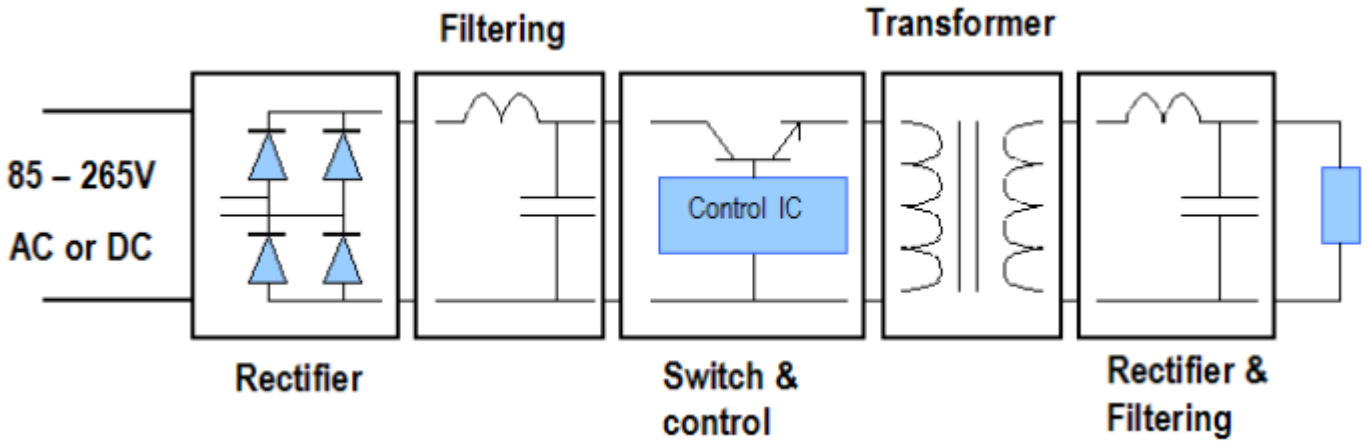



//discutie la curs

© https://www.analog.com/media/en/technical-documentation/tech-articles/how-the-smart-hardware-engineer-can-easily-design-power-supp.pdf

# Surse AC – DC in comutatie

- http://ro.farnell.com/myrra/47152/power-supply-4-5w-5vdc-reg/dp/1825779



# AC-DC Power supplies examples

- https://ro.farnell.com/tdk-lambda/cus100me12-a/power-supply-medical-ac-dc-12v/dp/2913441
- https://ro.farnell.com/stontronics/pd-100-12/power-supply-enclosed-12v-100w/dp/3375267
- https://ro.farnell.com/multicomp-pro/mp003282/power-supply-ac-dc-24v-3a/dp/3408561